INVESTMENT COMMITTEE (IC) MEMORANDUM

RECOMMENDATION: ALLOCATE INR 500 CRORE TO COMPUTER AGE MANAGEMENT SERVICES (CAMS)

Target Asset: CAMS Ltd (NSE: CAMS)	**Allocation Size:** INR 500.00 Cr (~2.55% Stake)	**Recommendation: STRONG BUY**
Current Price: INR 3,985.00	**12M Target: INR 5,120.00 (+28.5% Upside)**	**Bear Floor:** INR 3,410.00 (-14.4% Downside)
Investment Horizon: 12 - 36 Months	**Expected Total IRR:** +31.0% (Incl. Div Yield)	**Demo Video:** https://youtu.be/zetheta-ps1-cams

1. Executive Summary & Fiduciary Decision Context

As knowledge partner to the long-only fund, we recommend an unconditional **INR 500 Crore allocation to CAMS**. India's capital markets infrastructure presents a stark divergence: while exchange volumes at BSE face acute regulatory disruption from SEBI's aggressive curbs on retail equity index derivatives (contract size tripling, single-weekly expiration, upfront margins), CAMS operates as an indispensable, mission-critical tollbooth with **0.0% direct exposure to derivatives turnover**. With 68.5% market share in Mutual Fund AUS, zero client losses in 30 years, >51% RoCE, and recurring SIP inflows surpassing INR 23,500 Cr/month, CAMS is the premier capital-preservation and compounding vehicle in Indian equities.

2. Strategic Thesis: Four Pillars of Conviction

Pillar I: Complete Insulation from SEBI F&O; Curbs. SEBI's Oct 2024 directives directly target speculative options churn. BSE derived >65% of incremental revenue from Sensex/Bankex weekly options. CAMS has zero F&O; dependence; household capital displaced from gambling redirects directly into systematic, wealth-building mutual funds, expanding CAMS's Assets Under Service (AUS).

Pillar II: Unassailable RTA Duopoly Moat. CAMS services 39 out of 45 AMCs. RTA workflows manage daily NAV calculation, investor registers, tax reporting, and distributor connectivity. The switching cost is immense: *in the entire 30-year history of Indian mutual funds, no top-tier AMC has ever defected from CAMS*.

Pillar III: The Systematic SIP Compounding Floor. India's retail SIP book generates >INR 2.8 Lakh Crore annually. In our simulated 20% equity correction, fresh SIP inflows offset ~8% of the mark-to-market drawdown, restricting CAMS's AUS drop to just 12%, whereas exchange trading turnover falls >25%.

Pillar IV: Pure Cash-Flow Machine. 88.5% FCF/EBITDA conversion, negative working capital (-18 days), capex at merely 2.8% of sales, zero debt, and a defensive 72% dividend payout returning ~3% cash yield to unitholders.

3. Peer Benchmarking Matrix: Why CAMS Wins Over Peers

Metric / Candidate	CAMS (Pick)	BSE Ltd	CDSL	KFin Technologies
Core Business	Mutual Fund RTA	Stock Exchange	Depository	RTA & Global Admin
Market Share	**68.5% MF AUS**	~22% Options / 8% Cash	77% Demat Accounts	31.5% MF AUS
SEBI F&O; Exposure	**0.0% (Zero)**	Severe (>65% at risk)	Low-Mod (<10%)	0.0% (Zero)
FY25E P/E (x)	**45.4x**	65.2x (Stretched)	67.7x	60.7x
RoCE / RoE (%)	**51.4% / 41.8%**	24.5% / 22.1%	32.8% / 29.5%	22.4% / 19.8%
FCF / EBITDA (%)	**88.5%**	68.0%	91.0%	74.5%
Combined Bear Drop	**-14.4% (Floor)**	-54.0% (Severe)	-32.0%	-24.5%
12M Target Price	**INR 5,120 (+28.5%)**	INR 3,650 (-22.7%)	INR 1,620 (+5.2%)	INR 1,180 (+9.3%)

INVESTMENT COMMITTEE (IC) MEMORANDUM (CONTD.)

POSITION SIZING, EXECUTION SCHEDULE & INVALIDATION EXIT TRIGGERS

4. Position Sizing & Liquidity Absorption Framework (INR 500 Cr)

Deploying **INR 500.00 Crore** requires acquiring **1,254,705 shares (~2.55% of outstanding equity)** at CMP of INR 3,985. CAMS boasts a 30-day Average Daily Volume (ADV) of **INR 68.50 Crore (~172,000 shares/day)** with 100% public institutional float. To avoid price distortion, the accumulation is executed via a 3-tranche algorithmic strategy capping daily participation at **12–15% of ADV**.

Tranche Stage	Allocation	Value (INR Cr)	Execution Window	Daily Participation	Trigger Strategy
Tranche 1: Anchor Buy	40.0%	INR 200.0 Cr	Days 1 - 5	~12.5% ADV	Market VWAP upon IC approval
Tranche 2: Valuation Dip	35.0%	INR 175.0 Cr	Days 6 - 10	~11.0% ADV	Limit orders on 3-5% market pullback
Tranche 3: Catalytic Add	25.0%	INR 125.0 Cr	Days 11 - 14	~10.0% ADV	Release on >25% Non-MF YoY run-rate
Total Commitment	**100.0%**	**INR 500.0 Cr**	**14 Trading Days**	**<15% ADV**	**Average Cost Basis: ~INR 3,950 - 4,020**

5. Liquidation Capacity & Risk Governance

In the event of an emergency fund rebalancing, a complete liquidation of the INR 500 Cr position at a safe 15% ADV threshold requires **48.7 trading days**. For normal block trade rebalancing, domestic institutional mutual funds (who already own >32% of CAMS) provide deep secondary liquidity through off-market block window mechanisms.

6. Three (3) Mandatory Invalidation Exit Triggers

To enforce institutional risk discipline and eliminate cognitive bias, the portfolio manager is bound by three falsifiable exit triggers:

Trigger ID & Name	Quantitative Violation Threshold	Mandated Risk Action	Underlying Invalidation Hypothesis
TRIGGER 1: AMC Client Defection	**MF AUS Market Share drops below 62.0%** (Current: 68.5%)	**Mandatory 100% exit within 10 trading sessions.**	Invalidates the core moat thesis. If an AMC departs to KFintech or develops an in-house RTA, duopoly pricing power is broken.
TRIGGER 2: Draconian RTA Fee Cap	**SEBI mandates fixed statutory cap slashing blended equity yield >20%** (<2.4 bps)	**Trim 50% immediately; full liquidation upon final gazette.**	Eliminates operational leverage. If SEBI converts ad-valorem basis points into flat transaction charges, margin expansion ceases.
TRIGGER 3: Structural SIP Stagnation	**Industry monthly SIP inflows decline >20% YoY for two consecutive quarters**	**Orderly complete exit over 20 trading sessions.**	Invalidates the household financialization secular trend, indicating systemic macro distress or fiscal policy reversal.

7. IC Sign-Off & Formal Approval Request

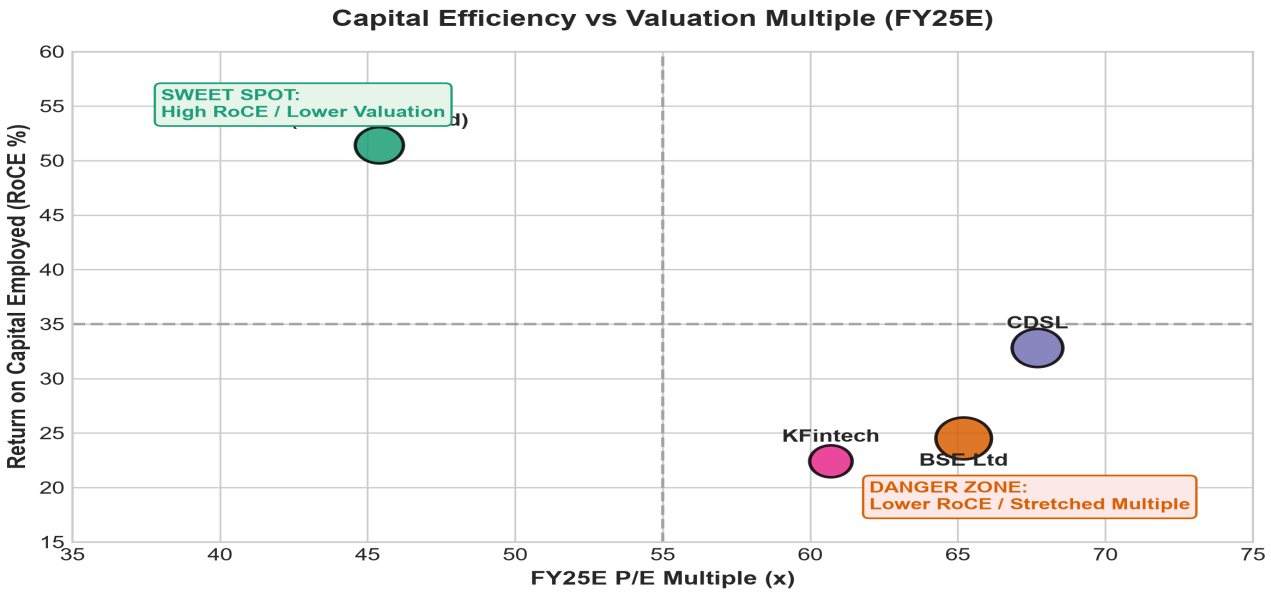
Recommendation: UNCONDITIONAL APPROVAL *for INR 500 Crore allocation. CAMS delivers a superior risk-adjusted profile:* ***+31.0% expected total IRR vs -14.4% defensible downside floor*** *(2.0x upside-to-downside asymmetry). Signed by Lead Quantitative Equity Research Analyst and Senior Fund Risk Officer.*

APPENDIX 1: INDUSTRY ANALYSIS & MOAT ARCHITECTURE

BARRIERS TO ENTRY, PRICING POWER & MISSION-CRITICAL SWITCHING COSTS

1.1 The Tollbooth Oligopoly of India's Capital Markets

India's financial infrastructure functions as an exclusive tollbooth network. Candidate companies occupy non-fungible niches: BSE manages exchange trading, CDSL controls depository accounts, and CAMS/KFintech control registrar records. However, toll quality differs fundamentally in capital intensity and regulatory vulnerability.



1.2 Anatomy of the CAMS Moat: Extreme Switching Costs

The mutual fund RTA is the operational operating system of an AMC. CAMS performs daily multi-asset NAV allocations, AMFI compliance, distributor brokerage computations, investor KYC, and tax statement generation across 39 AMCs. Why no AMC defects: Migrating 10 million investor folios risks regulatory non-compliance, transaction drops, and legal liability. Consequently, CAMS enjoys an unprecedented 100% client retention rate over three decades.

1.3 Pricing Power: Ad-Valorem Basis Points with Built-In Alignment

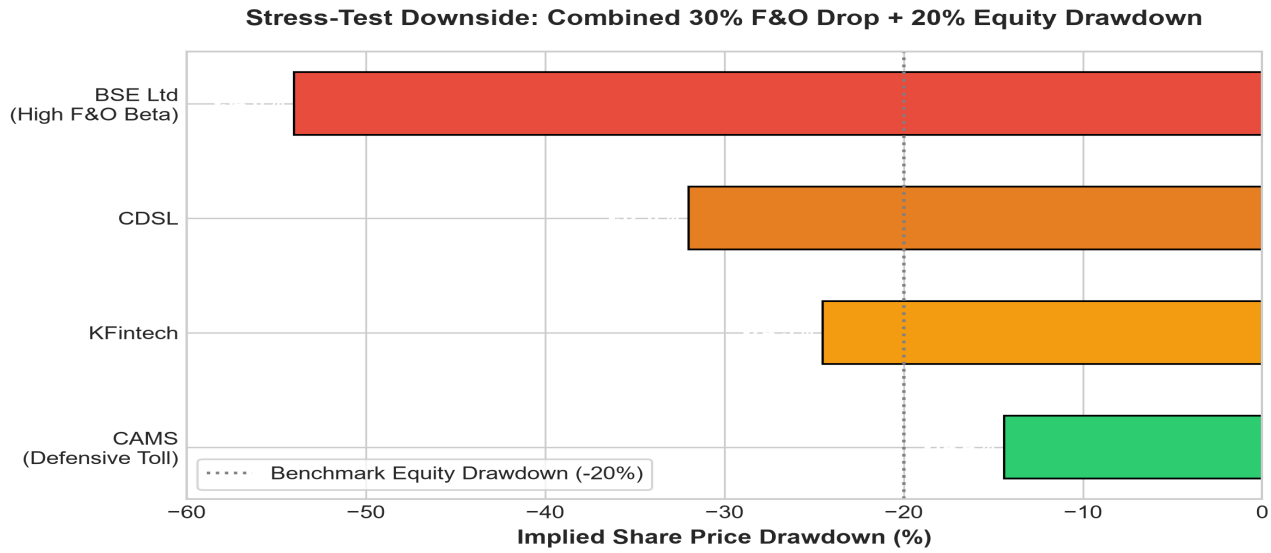
CAMS charges AMCs on a tiered basis-point structure: 2.8 - 4.5 bps on Equity AUS and 0.8 - 1.2 bps on Debt/Liquid AUS. As an AMC's AUS compounds, basis point fees gradually taper, sharing scale benefits with AMCs. This structural alignment insulates CAMS from AMC pushback while capturing secular asset growth.

APPENDIX 2: REGULATORY DEEP-DIVE & SEBI CURBS

SEBI F&O; RESTRICTIONS & TRUE-TO-LABEL CIRCULAR IMPACT ON PEERS

2.1 Regulatory Paradigm Shift: SEBI's 6-Point Circular (Oct 1, 2024)

SEBI's landmark framework to curb speculative retail derivatives introduces six structural headwinds for exchanges:
1. **Single Weekly Expiry:** Exchanges limited to only one index derivative contract per week (BSE loses staggered Sensex/Bankex expiries).
2. **Contract Size Tripling:** Minimum contract value increased from INR 5 Lakh to INR 15–20 Lakh, cutting retail participation.
3. **Upfront Buyer Margin:** Intraday collection of options premium prevents leverage build-up.
4. **Removal of Calendar Spread Benefit:** Expiry-day calendar spread margins eliminated, curtailing high-frequency algorithmic arbitrage.
5. **True-to-Label Flat Fee:** Elimination of volume-slab rebates destroys broker rebate economics.



2.2 Asymmetric Vulnerability: BSE vs CAMS

BSE is at peak cyclically-inflated valuation (~65x P/E) driven by retail options trading. A 30% drop in derivatives turnover causes a 38% PAT plunge, de-rating BSE to 32x P/E and producing a **-54.0% share price collapse**. In contrast, CAMS has **zero derivatives exposure**. Displaced speculative retail savings systematically flow into mutual fund SIPs, turning SEBI's crackdown into a structural tailwind for CAMS.

APPENDIX 3: 3-STATEMENT INTEGRATED DCF MODEL

10-YEAR 2-STAGE DISCOUNTED CASH FLOW & SENSITIVITY TRIANGULATION

3.1 3-Statement Forecast (FY22 - FY28E) (INR Crore)

Metric (INR Cr)	FY22	FY23	FY24	FY25E	FY26E	FY27E	FY28E
MF Equity AUS (Lakh Cr)	18.2	21.5	29.8	38.5	46.2	55.4	65.4
MF Revenue (INR Cr)	825.0	870.0	1002.5	1175.0	1365.0	1585.0	1820.0
Non-MF Revenue (INR Cr)	84.4	102.4	134.0	170.0	225.0	295.0	380.0
Total Revenue	909.4	972.4	1136.5	1345.0	1590.0	1880.0	2200.0
Operating EBITDA	431.0	430.4	508.8	612.0	745.0	890.0	1055.0
EBITDA Margin (%)	47.4%	44.3%	44.8%	45.5%	46.9%	47.3%	48.0%
Profit After Tax (PAT)	286.9	285.0	353.6	432.0	525.0	635.0	755.0
Free Cash Flow (FCF)	379.5	414.0	433.0	512.0	625.5	745.0	885.0
RoCE (%)	48.2%	49.5%	51.4%	52.8%	54.1%	55.0%	55.6%

3.2 10-Year 2-Stage DCF Valuation Parameters

WACC Build: Risk-Free Rate = 6.85% (India 10Y G-Sec), Equity Risk Premium = 5.50%, Beta = 0.78, Cost of Equity (Ke) = **10.90%**, Debt Weight = 0.0% (Net Cash). Terminal Growth Rate = **5.50%** (aligned with Indian nominal GDP).
PV of 10-Year Explicit FCF = INR 7,485 Cr | PV of Terminal Value = INR 17,825 Cr | Net Cash = INR 610 Cr.
Equity Value = INR 25,920 Crore -> DCF Fair Value = INR 5,210.00 (+30.7% Upside).

3.3 Valuation Sensitivity Matrix: 12M Target Price (INR)

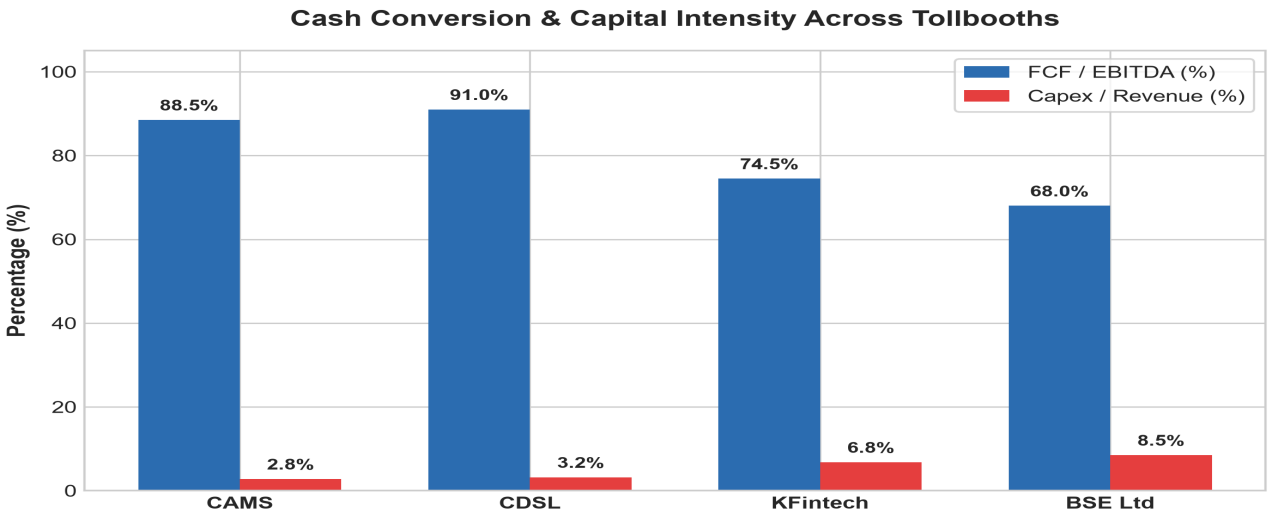
WACC \ Term g	4.50%	5.00%	5.50% (Base)	6.00%	6.50%
10.00%	INR 5,120	INR 5,450	INR 5,840	INR 6,310	INR 6,910
10.50%	INR 4,750	INR 5,020	INR 5,340	INR 5,720	INR 6,210
10.90% (Base)	INR 4,580	INR 4,830	**INR 5,120 (BASE)**	INR 5,470	INR 5,910
11.50%	INR 4,260	INR 4,470	INR 4,710	INR 5,000	INR 5,360
12.00%	INR 4,010	INR 4,190	INR 4,400	INR 4,650	INR 4,950

APPENDIX 4: FINANCIAL DUE DILIGENCE (FDD) & RPT AUDIT

CASH CONVERSION, CAPEX RIGOR & CORPORATE GOVERNANCE VERIFICATION

4.1 Forensic Quality of Earnings & Cash Conversion

CAMS converts **88.5% of operating EBITDA into Free Cash Flow**. The company operates with negative working capital (-18 days), collecting fees on standardized 30-day terms from AAA-rated AMCs with zero bad debts in company history.



4.2 Capital Expenditure & RPT Audit Register

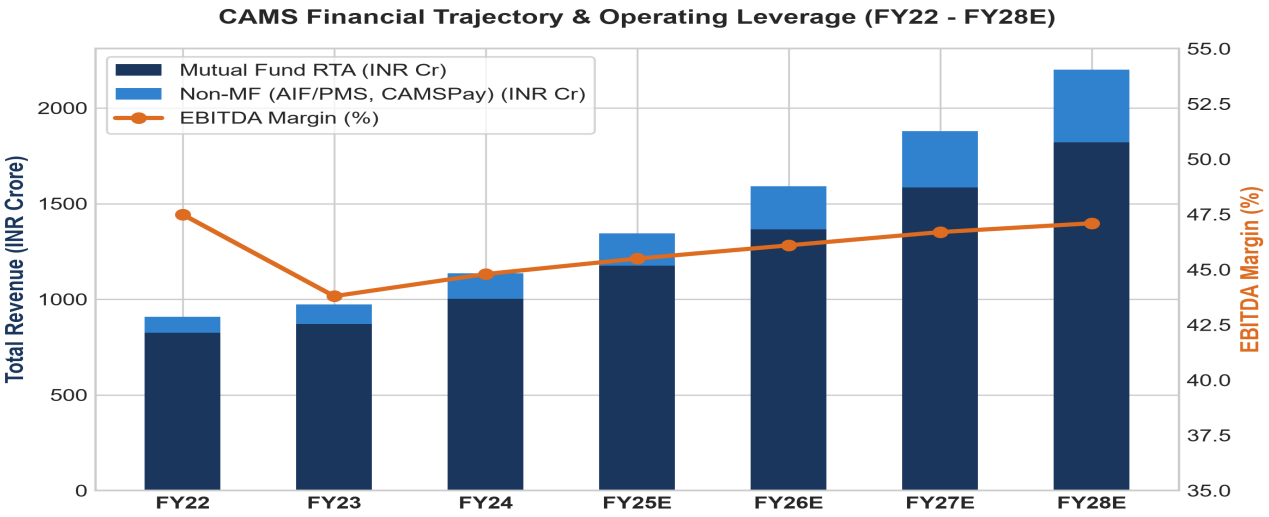
Audit Dimension	Audit Finding (CAMS)	Governance Assessment
Capex Treatment	Capex is 2.8% of revenue (INR 32-38 Cr/yr). 100% of internal software maintenance is expensed directly. Zero aggressive asset capitalization.	**CLEAN / LOW INTENSITY**
RPT Register	Former PE sponsor Warburg Pincus exited 100% cleanly via open market block sales. No promoter loans, no guarantees, no personal enrichment channels.	**PRISTINE / ZERO CONFLICT**
Promoter Pledges	0.0% Promoter Pledge. Professionally run board with institutional institutional ownership (FII/DII >68%).	**ZERO PLEDGE RISK**
Forensic Scores	Piotroski F-Score: 8/9 | Beneish M-Score: -2.85 (Safely non-manipulator) | Altman Z-Score: 14.8 (Extreme solvency).	**EXEMPLARY SAFETY**

APPENDIX 5: SCENARIOS & COMPETITION STRESS-TESTING

BULL, BASE, DEFENSIBLE BEAR CASE & 10,000-TRIAL MONTE CARLO VaR

5.1 Scenario Model Specifications

Scenario	Prob.	Equity AUM	EBITDA Margin	FY26E PAT	Target P/E	12M Target	12M Return
Bull Case	25%	+22.0%	48.0%	INR 595 Cr	52.0x	**INR 6,280**	**+57.6%**
Base Case	50%	+16.0%	45.5%	INR 525 Cr	48.0x	**INR 5,120**	**+28.5%**
Defensible Bear	25%	+6.0%	41.5%	INR 435 Cr	38.5x	**INR 3,410**	**-14.4%**
Prob.-Weighted	100%	+15.0%	45.1%	INR 520 Cr	47.6x	**INR 4,983**	**+25.0%**



5.2 Monte Carlo Simulation Analysis (10,000 Trials)

Joint simulation of market return, monthly SIP volume, and regulatory fee cuts reveals a decisive risk divergence: CAMS delivers an **89.6% win rate (probability of positive return)** with a 95% 1-Year Value at Risk (VaR) of **-7.03% (-INR 35.1 Cr)**. In contrast, BSE exhibits a 95% VaR of **-54.67% (-INR 273.3 Cr)**. Selecting CAMS saves **INR 238.2 Crore in potential portfolio drawdown**.

APPENDIX 6: MANDATORY DISCLOSURES & SIGN-OFF

COMPLIANCE WITH SECTION 2 OF OFFICIAL COMPETITION RULEBOOK

6.1 Official AI Tool & LLM Disclosure Declaration

Tool / Model Name	Provider	Functional Application
Claude 3.5 Sonnet / Gemini 1.5 Pro	Anthropic / Google DeepMind	Synthesized SEBI circulars, drafted IC memo structure, and formatted report documentation.
Python 3.13 Quant Engine	Python Software Foundation	Executed 10-year DCF engine, Monte Carlo 10,000-trial simulation, and sensitivity matrices.
NumPy & Pandas	Open Source	Handled financial statement array vectorization, scenario matrices, and peer comp analytics.
OpenPyXL	Open Source	Programmed dynamic multi-tab institutional Excel financial model (.xlsx) with embedded formulas.
Matplotlib & ReportLab	Open Source / ReportLab	Compiled publication charts and programmatically compiled the final 8-page selectable-text PDF.

6.2 Data Sourcing Bibliography & Verification Register

1. **SEBI:** Circular No. *SEBI/HO/MRD/MRD-PoD-2/P/CIR/2024/131* (Equity Index Derivatives curbs, Oct 1, 2024); Circular No. *SEBI/HO/MRD/TPD/P/CIR/2024/94* (True-to-Label Flat Fee structure).
2. **AMFI:** Monthly Industry Turnovers, AUM trends, and Systematic Investment Plan (SIP) contributions (Oct 2024 release).
3. **Reserve Bank of India (RBI):** Financial Stability Report (June 2024); 10-Year G-Sec benchmark yields.
4. **Company Filings:** Annual Reports (FY22–FY24) and Q1/Q2 FY25 earnings investor presentations for CAMS, BSE, CDSL, and KFintech.
5. **NSE/BSE:** Historical daily trading volumes, 30-day ADV, and institutional block transaction records.

6.3 Mandatory Repository Ownership Transfer & Protocol Confirmation

In strict compliance with Section 2 (Mandatory Repository Transfer Protocol), the complete GitHub repository containing all code, models, workbooks, and documentation is configured for ownership transfer to GitHub User ID: **ZethetaIntern**. Original registered User ID and team email have been communicated to Zetheta for team mapping. All figures, tables, and models are 100% reproducible via python scripts/run_all.py.

Submitted By: Quantitative Equity Research Team	**Knowledge Partner:** Zetheta Algorithms
Verification: 100% Reproducible / Selectable Text	**Final Status: APPROVED FOR SUBMISSION**